

Regime-Aware Spectral Risk Estimation for Cost-Constrained Statistical Arbitrage

Your Name

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Abstract

This paper studies whether random matrix theory (RMT) covariance regularization improves the risk allocation of a cost-constrained statistical arbitrage strategy. The proposed strategy separates the return signal from the risk model: a pre-specified residual mean-reversion signal determines expected returns, while spectral covariance estimates determine portfolio risk. We propose a regime-aware estimator that blends long- and short-window cleaned covariance matrices using a rolling measure of correlation drift. The estimator's mathematical validity and empirical value are separate questions. We establish positive semidefiniteness of the blend and specify a walk-forward design to test whether it improves out-of-sample risk and net performance. No empirical result is claimed in this draft.

1 Research question and contribution

The central question is whether adapting the covariance risk model to detected changes in correlation structure improves a statistical arbitrage portfolio after transaction costs.

The proposed extension is a regime-aware blend of long- and short-window RMT covariance estimates. The blend is designed to retain the stability of a longer estimation window while responding to changes indicated by a shorter window. Its usefulness is an empirical hypothesis, not an assumed result. The literature review must establish whether this exact construction and evaluation have already been studied.

2 Signal and portfolio model

Let $r_t \in \mathbb{R}^N$ denote the vector of asset returns at time t . A separate signal model produces a forecast $\hat{\mu}_t \in \mathbb{R}^N$. For example, asset returns may first be decomposed into factor and residual components:

$$r_t = Bf_t + \varepsilon_t,$$

where f_t is a vector of common factors, B is a factor loading matrix, and ε_t contains residual returns.

A possible mean-reversion signal uses standardized residuals:

$$z_{i,t} = \frac{\hat{\varepsilon}_{i,t}}{\hat{s}_{i,t}}, \quad \hat{\mu}_{i,t} = -\theta_i z_{i,t}.$$

The factor model, residual scale, and parameters θ_i must be estimated using information available at time t only. This signal specification is a starting point; its predictive value must be tested independently of the covariance estimator.

Given a covariance estimate $\widehat{\Sigma}_t$, portfolio weights are chosen by solving

$$\widehat{w}_t = \arg \max_{w \in \mathcal{W}} \left\{ \widehat{\mu}_t^\top w - \frac{\lambda}{2} w^\top \widehat{\Sigma}_t w - c \|w - w_{t-1}\|_1 \right\}.$$

Here $\lambda > 0$ controls risk aversion, $c \geq 0$ represents proportional trading costs, and $\mathcal{W}$ contains leverage, position, and exposure constraints. These choices must be fixed or calibrated using training data, not the final test period.

3 Spectral covariance estimation

Let $X \in \mathbb{R}^{T \times N}$ be a rolling matrix of demeaned, standardized returns, with T observations and N assets. The sample correlation matrix is

$$C = \frac{1}{T} X^\top X.$$

Write its eigendecomposition as

$$C = U \text{diag}(\lambda_1, \dots, \lambda_N) U^\top.$$

Under an idealized null model of independent standardized observations, the Marchenko–Pastur distribution describes the limiting eigenvalue spectrum as $N, T \rightarrow \infty$ with $q = N/T$ fixed. For unit population variance, the bulk support is

$$\lambda_- = (1 - \sqrt{q})^2, \quad \lambda_+ = (1 + \sqrt{q})^2.$$

This null model is a reference, not a complete model of financial returns. Serial dependence, heavy tails, and changing correlations can affect the spectrum and must be considered in robustness analysis.

As a baseline cleaning rule, eigenvalues inside the estimated noise bulk may be replaced by their average, while selected eigenvalues above the bulk are retained. The cleaned correlation matrix is reconstructed from the original eigenvectors and transformed eigenvalues. Marginal volatilities are then restored to obtain a covariance estimate.

4 Proposed regime-aware extension

At each time t , construct cleaned correlation estimates from long and short rolling windows, denoted by $\widetilde{C}_t^{(L)}$ and $\widetilde{C}_t^{(S)}$. Define their relative drift as

$$d_t = \frac{\|\widetilde{C}_t^{(S)} - \widetilde{C}_t^{(L)}\|_F}{\|\widetilde{C}_t^{(L)}\|_F + \epsilon},$$

where $\epsilon > 0$ prevents division by zero. For a scale parameter $\tau > 0$, set

$$\alpha_t = \frac{d_t}{d_t + \tau}, \quad \widetilde{C}_t^{(A)} = (1 - \alpha_t) \widetilde{C}_t^{(L)} + \alpha_t \widetilde{C}_t^{(S)}.$$

The parameter τ , window lengths, and all preprocessing choices must be selected using training data only. The study will compare this rule with fixed-window RMT cleaning and established shrinkage baselines.

Proposition 1. *If $\tilde{C}_t^{(L)}$ and $\tilde{C}_t^{(S)}$ are positive semidefinite and $0 \leq \alpha_t \leq 1$, then $\tilde{C}_t^{(A)}$ is positive semidefinite.*

Proof. For any vector $x \in \mathbb{R}^N$,

$$x^\top \tilde{C}_t^{(A)} x = (1 - \alpha_t) x^\top \tilde{C}_t^{(L)} x + \alpha_t x^\top \tilde{C}_t^{(S)} x.$$

Both quadratic forms on the right are nonnegative, and both coefficients are nonnegative. Therefore $x^\top \tilde{C}_t^{(A)} x \geq 0$ for every x , which proves the claim. $\square$

This proposition establishes mathematical validity of the blend. It does not establish that the estimator is more accurate or profitable.

5 Empirical hypotheses

The walk-forward study will test the following hypotheses:

- H1. The regime-aware estimate reduces realized out-of-sample portfolio variance relative to the sample covariance baseline.
- H2. Any risk improvement persists after turnover and transaction costs.
- H3. The estimated benefit is robust to reasonable changes in window lengths, market periods, and cost assumptions.

These are hypotheses to test, not conclusions.

6 Evaluation design

All preprocessing, universe membership, parameter selection, signal estimation, and portfolio construction must use only information available at each decision date. Use nested walk-forward evaluation: an inner training period selects parameters, and a later untouched period evaluates them.

Baselines should include equal weighting, sample covariance, a standard linear-shrinkage estimator, fixed-window RMT cleaning, and the proposed regime-aware blend. Report realized volatility, drawdown, turnover, gross and net performance, and uncertainty across time periods. Include transaction costs and short-borrow assumptions where applicable.

The final study should disclose data sources, survivorship and look-ahead controls, missing-data rules, rebalance timing, and all parameter choices.

7 Results

To be completed after the empirical protocol is fixed and the experiments are run. Do not insert performance claims before then.

8 Limitations

The Marchenko–Pastur benchmark assumes an idealized sampling model. Financial returns can violate its assumptions through serial dependence, heavy tails, heteroskedasticity, and structural breaks. The proposed drift statistic can also react to estimation noise. These issues motivate robustness checks; they do not guarantee that the proposed estimator will outperform its baselines.

References

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